DAILY CASH POSITION

Aim

To write a COBOL program to calculate the closing cash balance of a bank branch.

Algorithm

1. Start.
2. Read opening cash, cash received, and cash paid.
3. Add opening cash and cash received.
4. Check whether cash paid is greater than available cash.
5. If it is greater, display CASH RECONCILIATION ERROR.
6. Otherwise, subtract cash paid from available cash and display the closing cash.
7. Stop.

COBOL Program

```
IDENTIFICATION DIVISION.
PROGRAM-ID. CASH-POSITION.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 OPEN-CASH       PIC 9(9).
01 CASH-RECEIVED   PIC 9(9).
01 CASH-PAID       PIC 9(9).
01 TOTAL-CASH      PIC 9(10).
01 CLOSE-CASH      PIC Z(9)9.

PROCEDURE DIVISION.
   ACCEPT OPEN-CASH
   ACCEPT CASH-RECEIVED
   ACCEPT CASH-PAID
```

```
COMPUTE TOTAL-CASH = OPEN-CASH + CASH-RECEIVED


IF CASH-PAID > TOTAL-CASH
   DISPLAY "CASH RECONCILIATION ERROR"
ELSE
   COMPUTE CLOSE-CASH = TOTAL-CASH - CASH-PAID
   DISPLAY "Closing cash: " CLOSE-CASH
END-IF


STOP RUN.
```

Sample Input

50000

12000

45000

Output

Closing cash: 17000

Approach

First, I understood that the problem is about calculating the closing cash balance of a bank branch at the end of the day.

1. **Identify the inputs:** I identified three inputs: opening cash, cash received, and cash paid out.
2. **Calculate total cash:** I added the opening cash and cash received to determine the total available cash.
3. **Validate the payment:** Before calculating the closing balance, I checked whether the cash paid out exceeded the total available cash.
4. **Handle the condition:** If the cash paid out is greater than the total cash, I display “CASH RECONCILIATION ERROR.”

5. **Calculate the closing balance:** Otherwise, I subtract the cash paid out from the total cash and display the closing balance.

Formula:

Total Cash = Opening Cash + Cash Received

Closing Cash = Total Cash − Cash Paid Out

Example:

Opening Cash = 50000
Cash Received = 12000
Cash Paid Out = 45000

Total Cash = 50000 + 12000 = 62000

Closing Cash = 62000 − 45000 = 17000

Therefore, the closing cash balance is **17000**.

Result

Thus, the COBOL program calculates the closing cash balance as ₹17,000. If the cash paid exceeds the available cash, it displays CASH RECONCILIATION ERROR.